TAPFormer: Robust Arbitrary Point Tracking via Transient Asynchronous Fusion of Frames and Events

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CVPR
JUNE 3-7, 2026
DENVER
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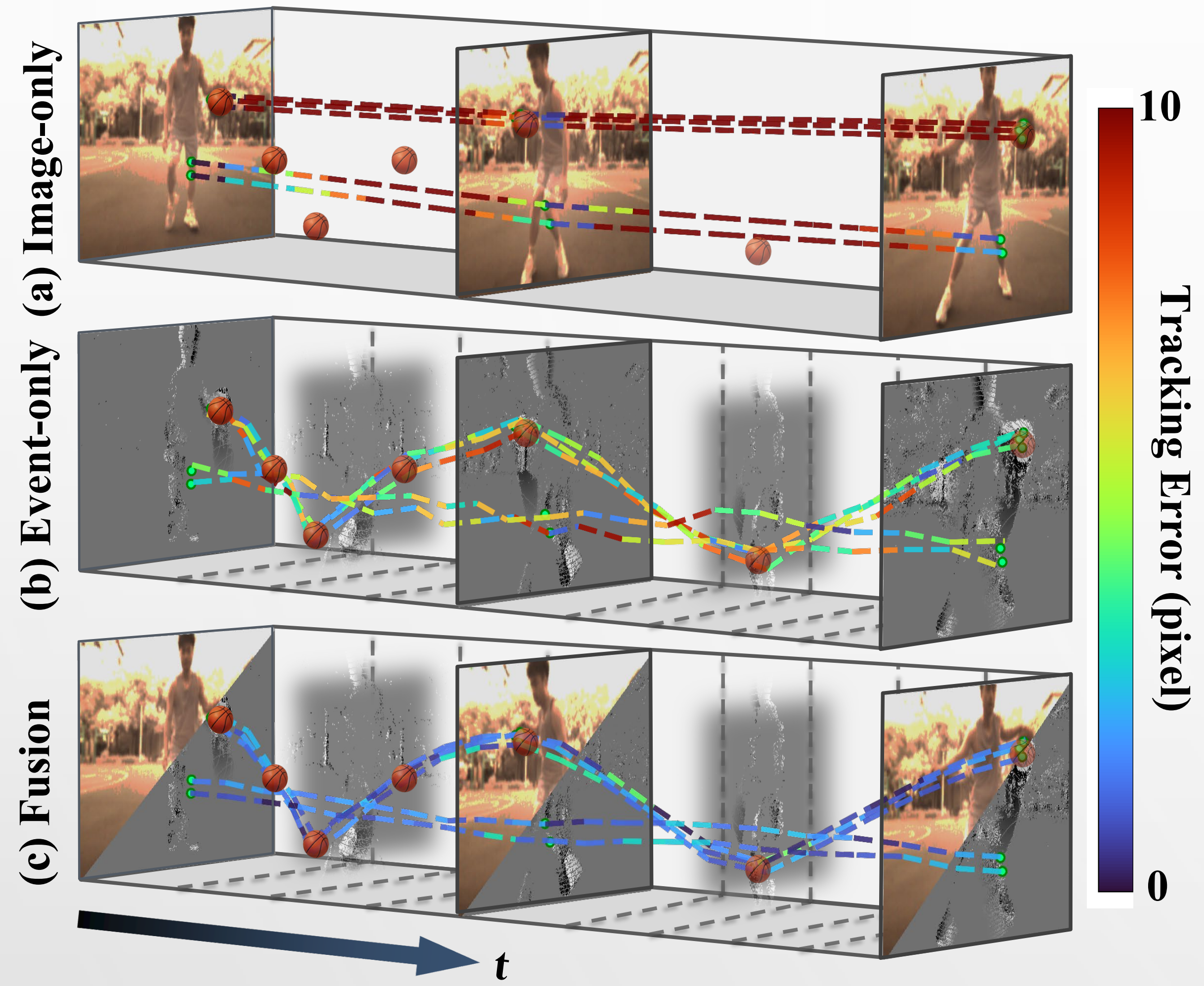
Motivation



- **Image-only tracking methods** degrade rapidly or fail under challenging conditions like rapid motion, drastic lighting changes, and limited frame rates.
- **Event-only tracking methods** struggle with high-precision tracking due to motion-coupled feature instability, and can even fail completely in the absence of relative motion.
- **Our goal** is to achieve robust arbitrary point tracking across diverse real-world scenarios, from fast to slow motion and from daytime to nighttime, by fusing image and event information.

Main Contribution

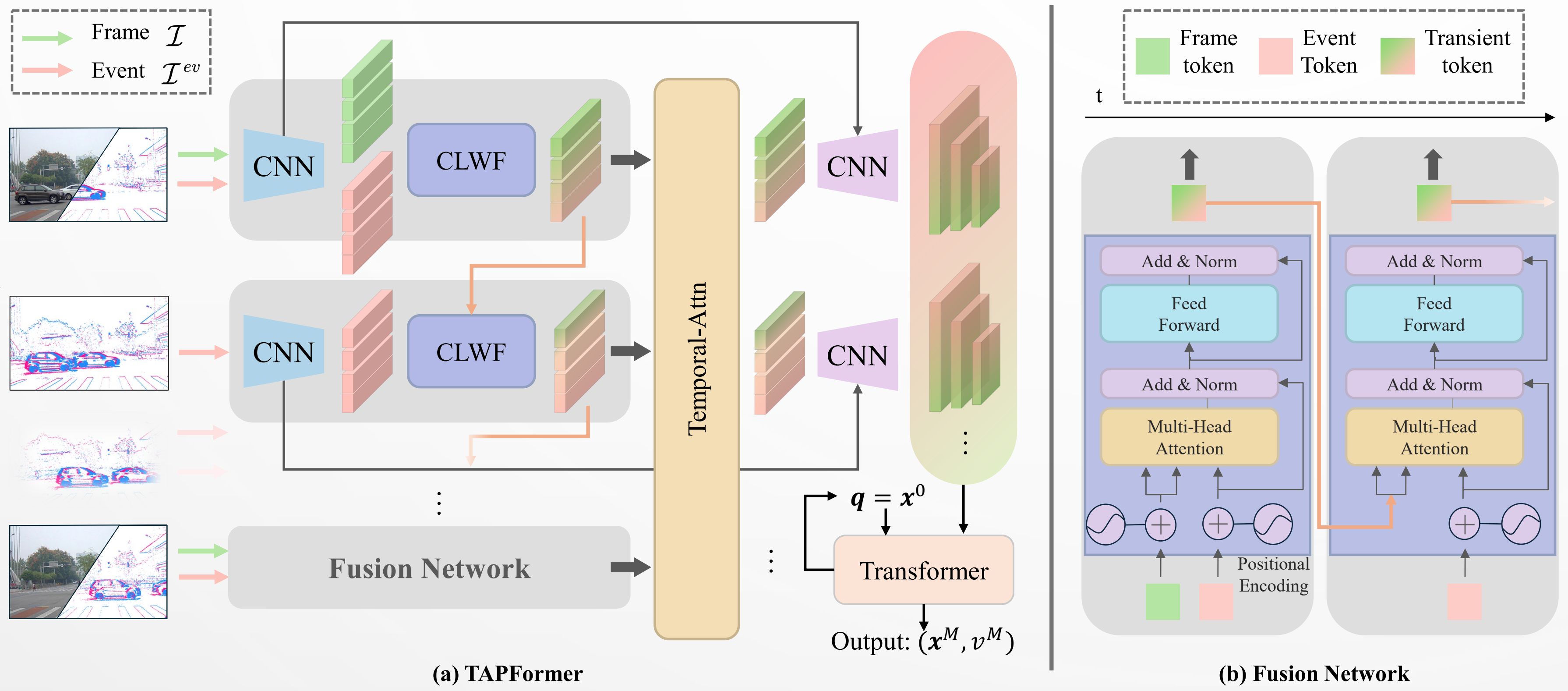
◆ We propose a unified framework that fully exploits the complementary strengths of RGB frames and event streams, enabling robust arbitrary point tracking under diverse challenging real-world conditions.

◆ The first real-world TAP benchmark covering challenging conditions with synchronized frame–event data.

◆ State-of-the-art performance across multiple datasets in TAP and feature point tracking tasks, demonstrating the effectiveness and generality of our approach.

Motivation

We propose an asynchronous fusion framework with two key modules: a Transient Feature Update Mechanism that bridges sensor frequency mismatch for high-frequency tracking, and a Cross-Modal Locally Weighted Fusion Module that adaptively selects reliable local regions. Together, they ensure robust arbitrary point tracking under blur, occlusion, and illumination variations.



(a) TAPFormer (b) Fusion Network

Problem Definition:
our goal is to use high-frequency event representations $(\mathcal{I}^{ev})$, the images $(\mathcal{I})$, and the initial query state q, to estimate the pixel trajectory (x_{τ_t}) and visibility state (v_{τ_t}) of the query point at each (τ_t, about 200Hz).

Mathematical Formulation:

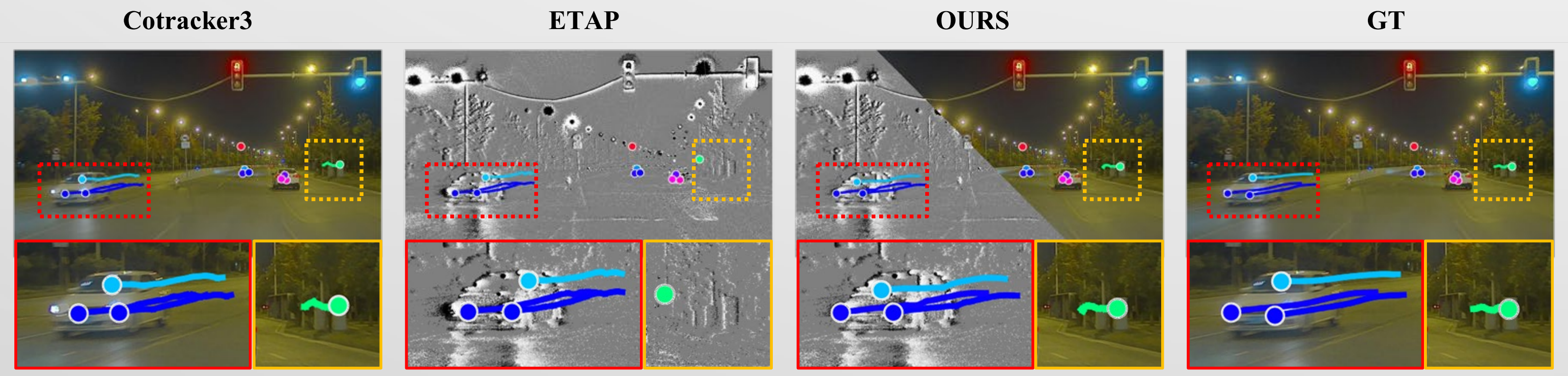
$$\mathcal{R}_t = \mathcal{G}(\Phi_I(I_t), \Phi_E(\mathcal{F}(\mathcal{E} \cap \mathcal{W}_t)))$$

$$\mathcal{R}_{t+\Delta} \leftarrow \mathcal{U}(\mathcal{R}_{t+\Delta-1}, \Phi_E(\mathcal{F}(E_{t+\Delta})))$$

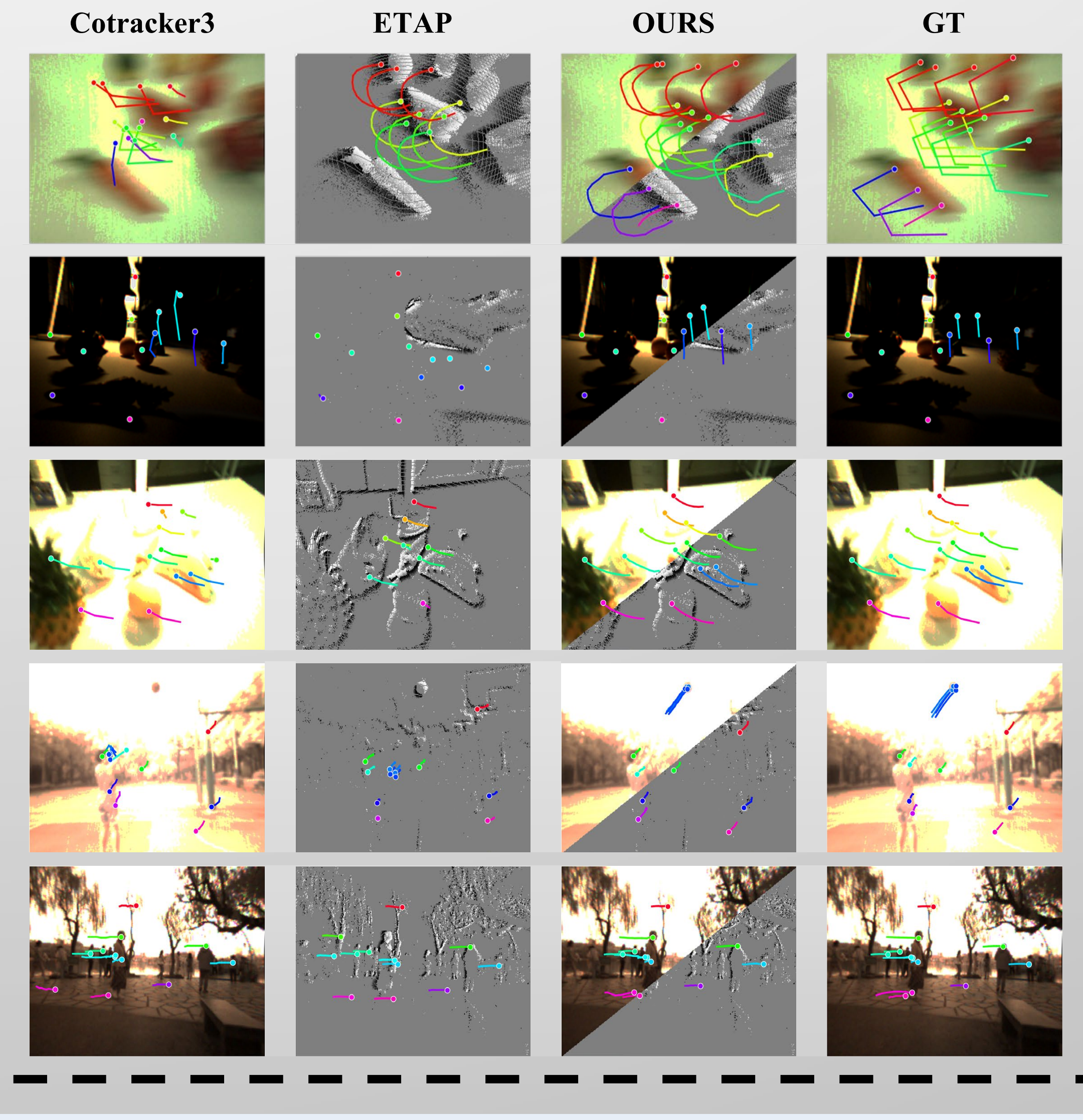
$\mathcal{R} \rightarrow Transient\ token;\ \mathcal{F}(\cdot) \rightarrow SBT\ eventrepresentations;$
$\mathcal{G}(\cdot) \rightarrow CLWF\ module;\ \mathcal{U}(\cdot) \rightarrow Update\ module;$
$\Phi(\cdot) \rightarrow modality - specificencoders;$

Experiments

We compared our method against various state-of-the-art algorithms on our self-collected real-world multimodal arbitrary point tracking datasets, InivTAP and DrivTAP.

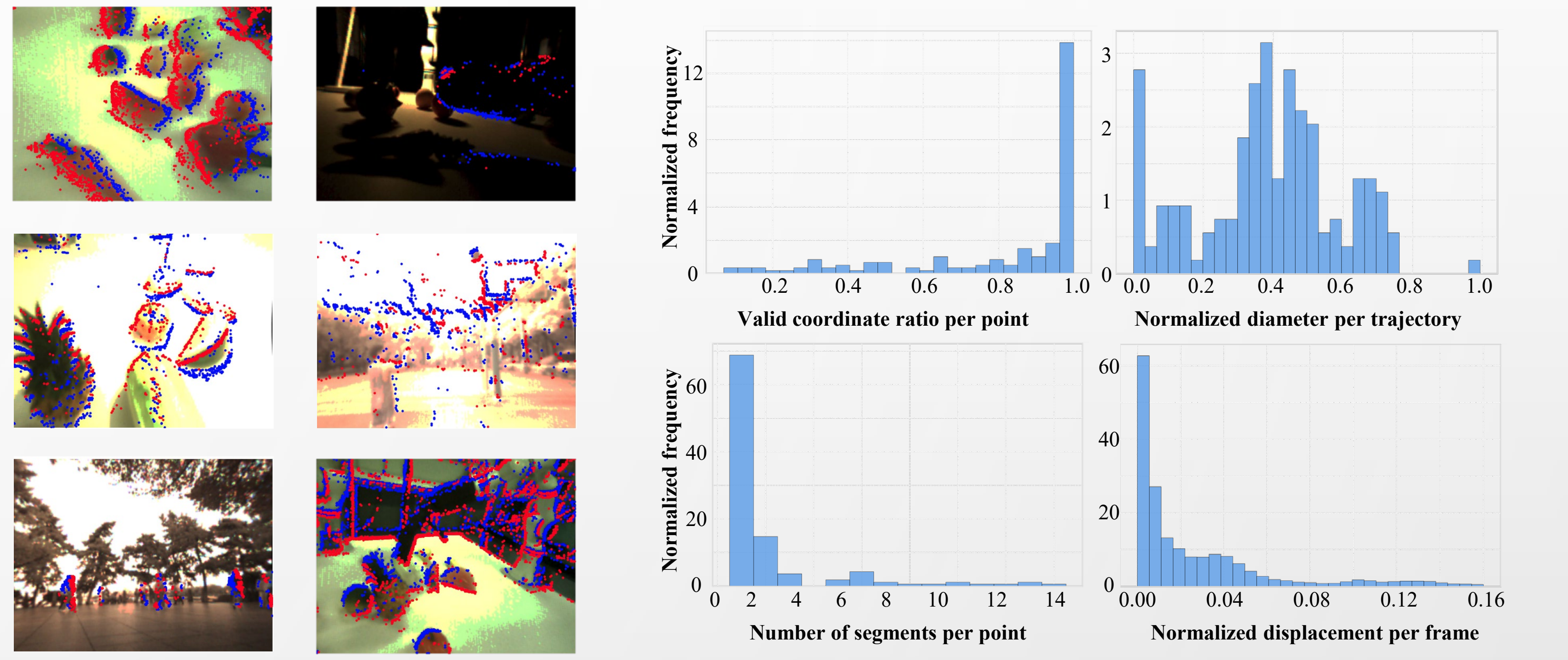


Method	Train Data	Input	Time↓	InivTAP			DrivTAP			Mean δ^{vis}_{avg} ↑
				AJ↑	δ^{vis}_{avg} ↑	OA↑	AJ↑	δ^{vis}_{avg} ↑	OA↑	
PIPs++ [57]	PO	Frame	21.1	–	31.2	–	–	37.8	–	34.5
Chrono [26]	Kub	Frame	-	31.2	44.7	79.8	15.8	30.8	68.1	37.8
CoTracker3 [25]	Kub	Frame	19.0	41.8	53.2	72.8	37.1	46.5	95.4	49.9
TAPFormer-F (Ours)	FastKub	Frame	16.5	44.6	54.5	74.7	36.6	46.7	93.8	50.6
E2Vid [43]-PIPs++ [57]	PO	Event	21.4	–	14.3	–	–	30.4	–	22.4
E2Vid [43]-Chrono [26]	Kub	Event	-	10.4	17.9	68.8	8.1	16.9	68.7	17.4
E2Vid [43]-CoTracker3 [25]	Kub	Event	22.2	12.8	19.4	56.6	18.6	27.0	89.4	23.2
ETAP [16]	E-Kub	Event	36.9	12.8	22.3	86.3	13.5	27.8	68.1	25.1
TAPFormer-E (Ours)	E-FastKub	Event	**14.0**	20.9	29.8	79.0	28.9	37.6	90.2	33.7
FETAP [33]	MF	Frame+Event	-	–	25.7	–	–	30.3	–	28.0
FETAP [33]	FE-FastKub	Frame+Event	23.4	42.2	54.9	83.9	36.8	46.4	95.2	50.7
TAPFormer (Ours)	FE-FastKub	Frame+Event	20.2	**57.0**	**69.9**	**95.2**	**48.8**	**60.1**	**97.8**	**65.0**



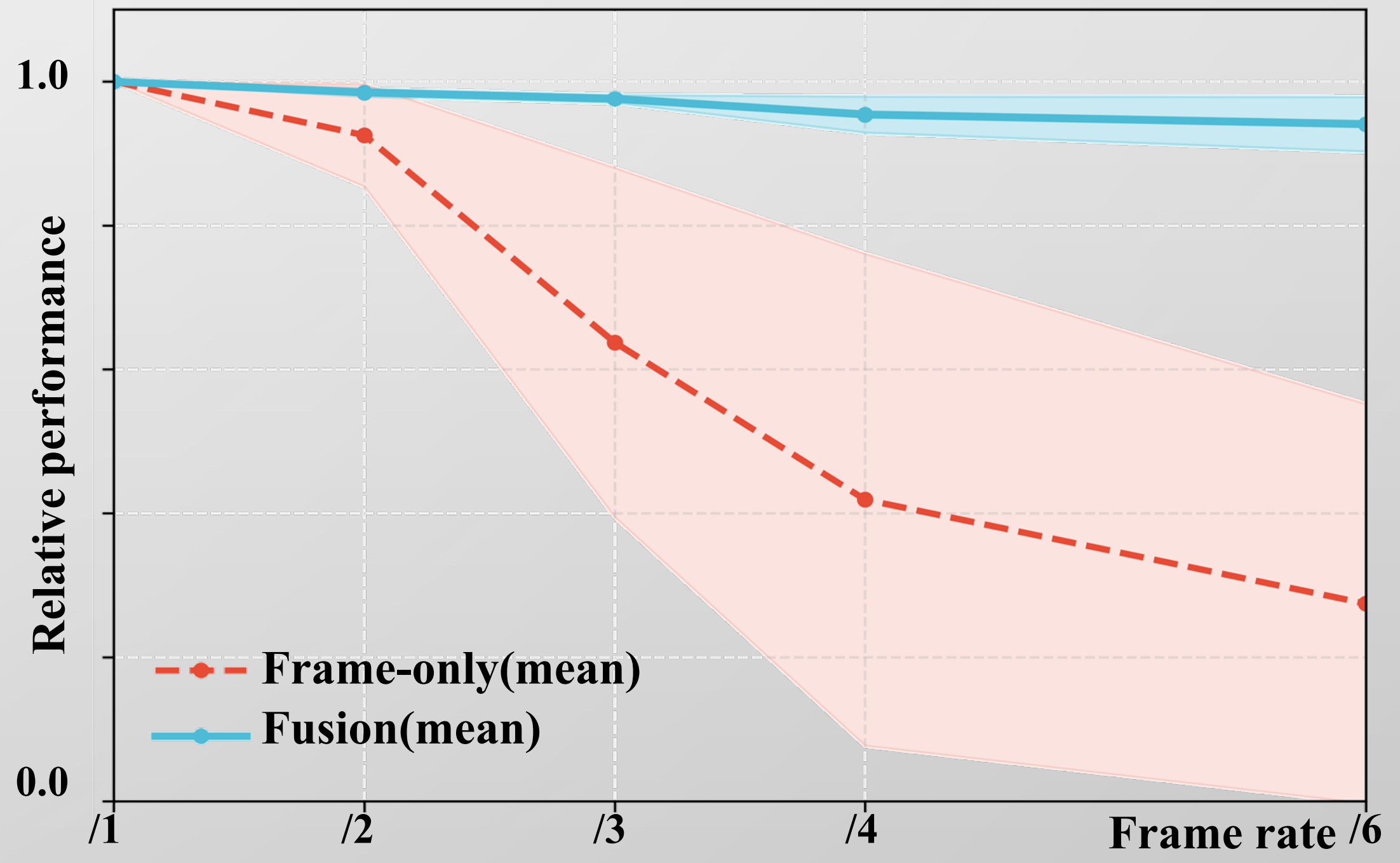
Datasets

We propose the first real-world synchronized frame-event TAP dataset, annotated using optical flow-assisted manual calibration.

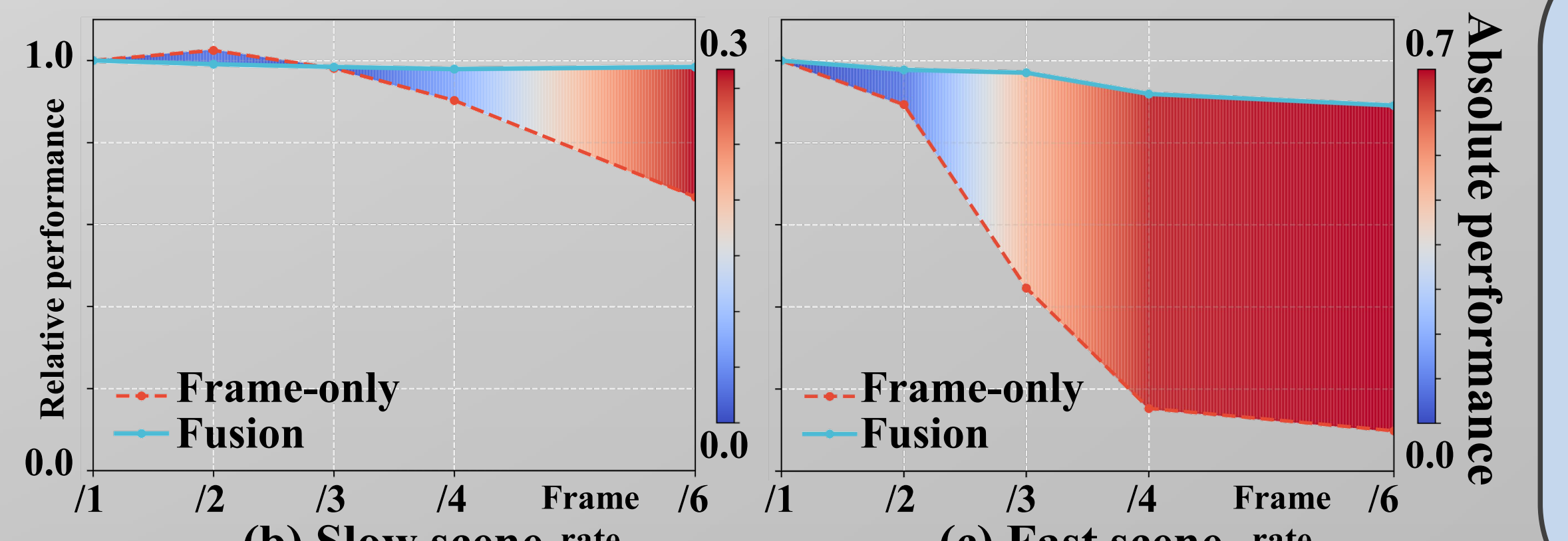


More Analysis

Frame-Rate Analysis

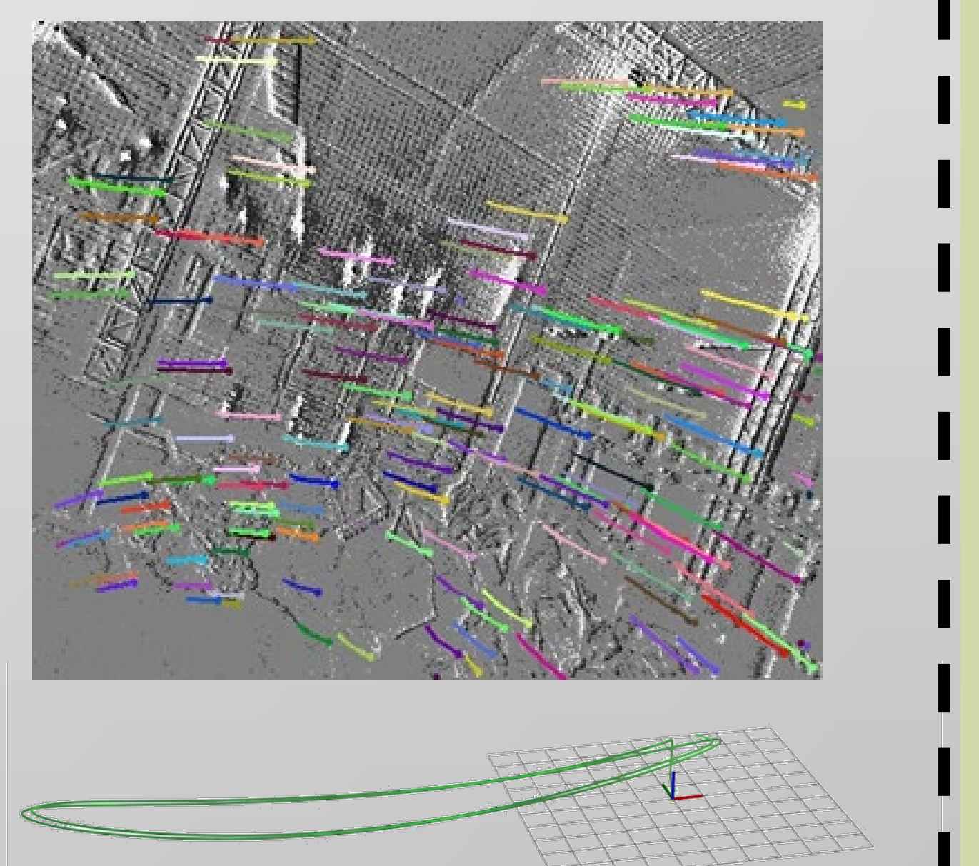


(a) Average performance with standard deviation



(b) Slow scene (c) Fast scene

SLAM Applications



Scan for more visualizations, code, and datasets—now fully open-sourced!

Project homepage